

Viscerale Smerter

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Læringsmål

- Redegør for forskel mellem smerter af somatisk eller visceral oprindelse
- Forklar generelle kendetegn ved mavesmerter, som smertetype, varighed og medvirkende udløsende faktorer
- Relater placering, kvalitet og henvisning af mavesmerter til den underliggende patologi
- Foretag en hensigtsmæssig undersøgelse af en patient med mavesmerter

Smertetyper

Somatiske nociceptive (± inflammation)
Viscerale nociceptive (± inflammation)
Neuropatiske / neurogene
- Skade på nerver / nervebaner

Nociplastisk
- sensitivt smertesystem (øget følsomhed)

Akutte versus kroniske



Nociception vs. smerte



- Nociception: neural aktivitet
  - Ubevidst sensorik, der foregår perifert eller central gennem en aktivering af sensoriske transducere (receptorer). Eks. homeostatisk regulering af mæthed
- Smerte: “Ubehagelig sensorisk og emotional oplevelse, som er knyttet til – eller ligner det du oplever – vævsskade eller truende vævsskade”
  - Involverer kognitive og affektive komponenter
  - ”No brain – no pain”

Og mere…

Smerter er altid en personlig oplevelse, der i forskellig grad kan påvirkes af biologiske, psykologiske og sociale faktorer.

Smerter og nociception er to forskellige ting.

Smerter kan ikke udelukkende knyttes til aktivitet i sensoriske neuroner.

Det er gennem oplevelser i livet, at enkeltpersoner lærer, hvad smerte er.

En persons beskrivelse af oplevede smerter skal respekteres.

Selv om smerte normalt er en del af kroppens beskyttelses-mekanismer og tilpasning, kan smerter have negative effekter på funktionel evne og social såvel som psykologisk velvære.

Somatisk vs. viserale smerter

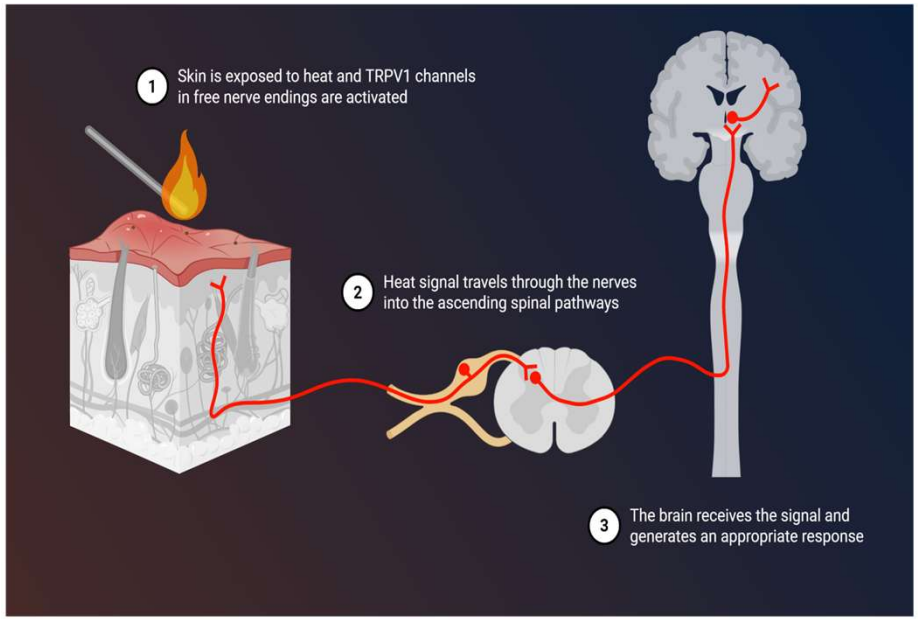
Somatic	Visceral
• Nociceptors are involved • Often well localized • Usually described as throbbing or aching • Can be superficial (skin, muscle) or deep (joints, tendons, bones)	• Involves hollow organ and smooth muscle nociceptors that are sensitive to stretching, hypoxia and inflammation • Pain is usually referred, poorly localized, vague and diffuse • May be associated with autonomic symptoms (e.g., pallor, sweating, nausea, blood pressure and heart rate changes)

Diffuse/poor localization:
- low density of afferent innervation
- Wide divergence of visceral nociceptive pathways in CNS

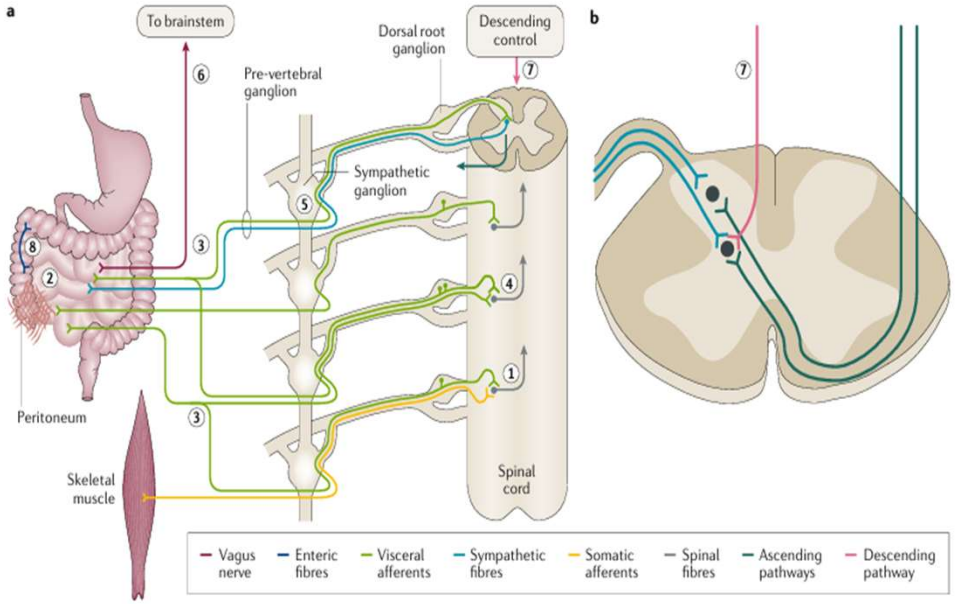
Poor discrimination
- Convergence of visceral and somatic nociceptive afferents (spinal cord)

McMahon SB, Koltzenburg M. Wall and Melzack´s Textbook of Pain. 5th ed. Elsevier; London, UK; 2006.
Sikandar S, Dickenson SH. Curr Opin Support Palliat Care 2012; 6(1): 17-26.

Somatic vs. visceral



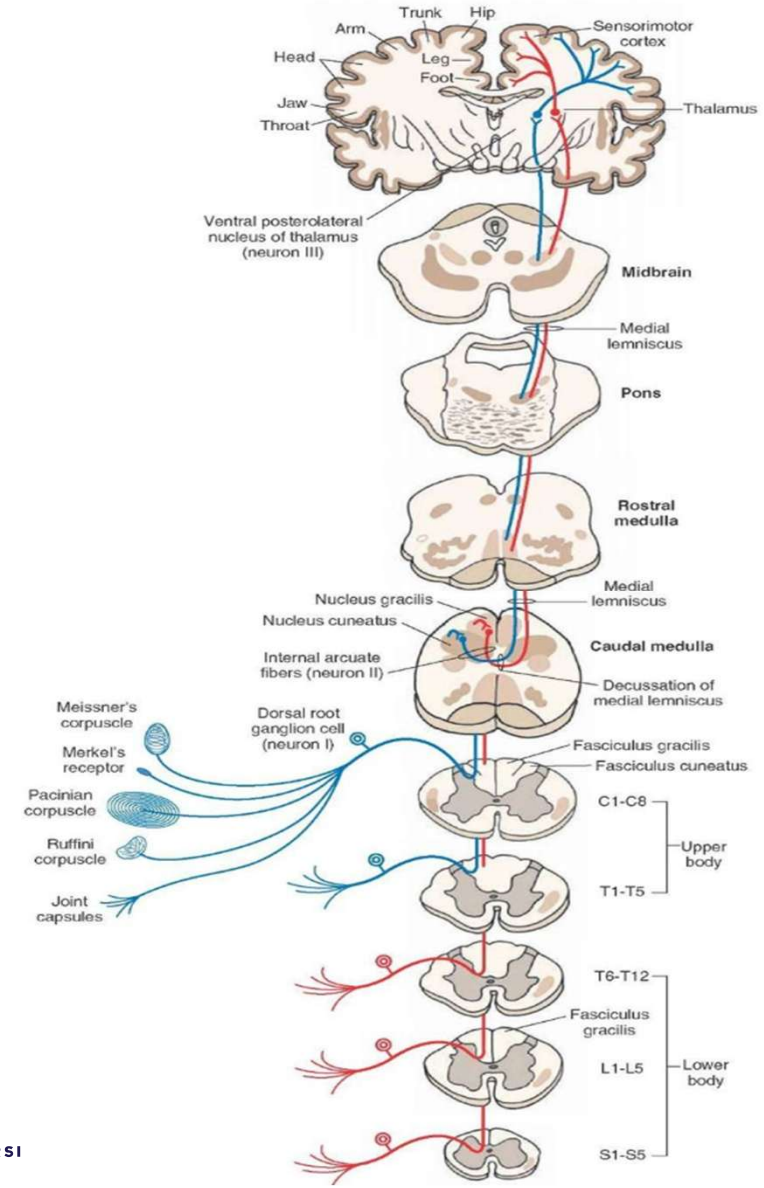
Aδ- and C-fibers ~ equally represented
More non-peptidergic fibers (different neurochemistry, targets and functions)



Aδ-fibers ~ 40%, more C-fibers + polymodal interoceptors (respond to non-damaging and damaging stimuli) + larger amount of "silent" nociceptors
ENS: "second brain" (intrinsic)

Ascenderende transmission

- Visceral afferents via dorsal horn, synapsing in different lamina
- Transmission gennem medulla, hjernestamme, den forlængede rygmarv, pons, midthjerne til thalamus (fungerer som relæ station)

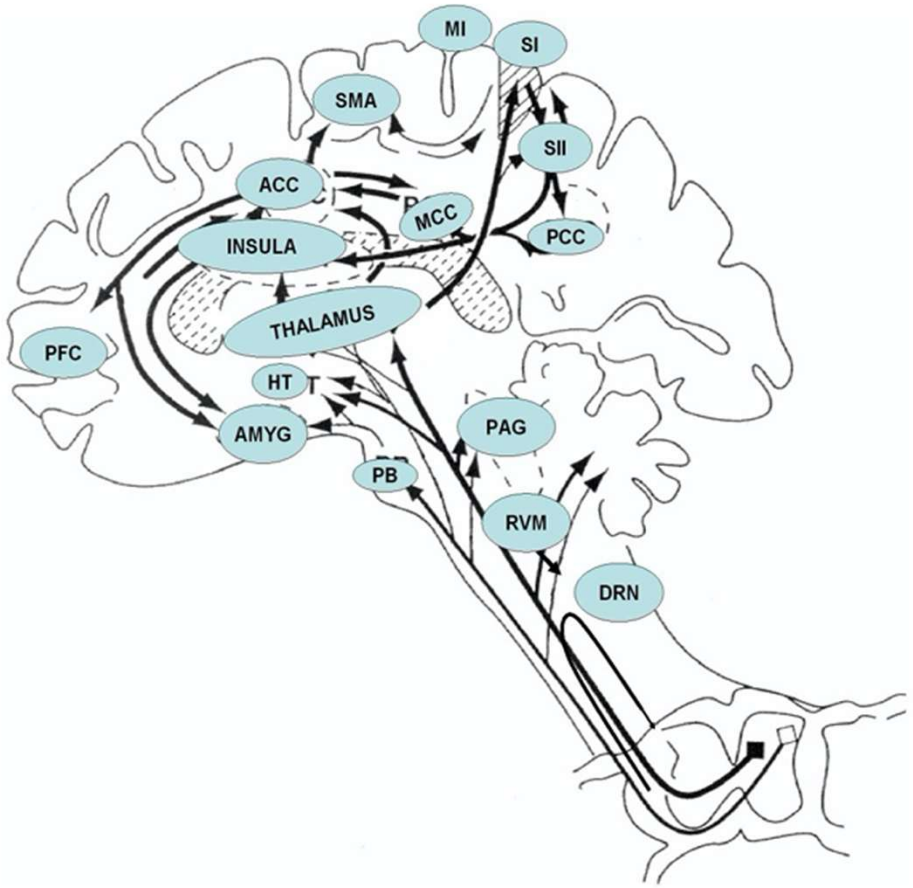


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Meerschaert and Chiu 2025.
https://doi.org/10.1038/s41575-024-01017-9

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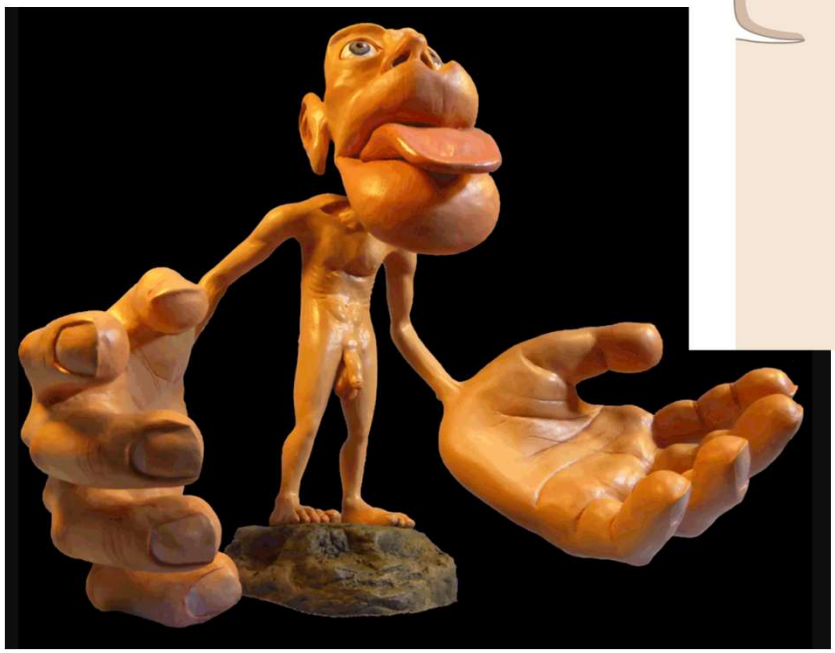
Supraspinal kommunikation

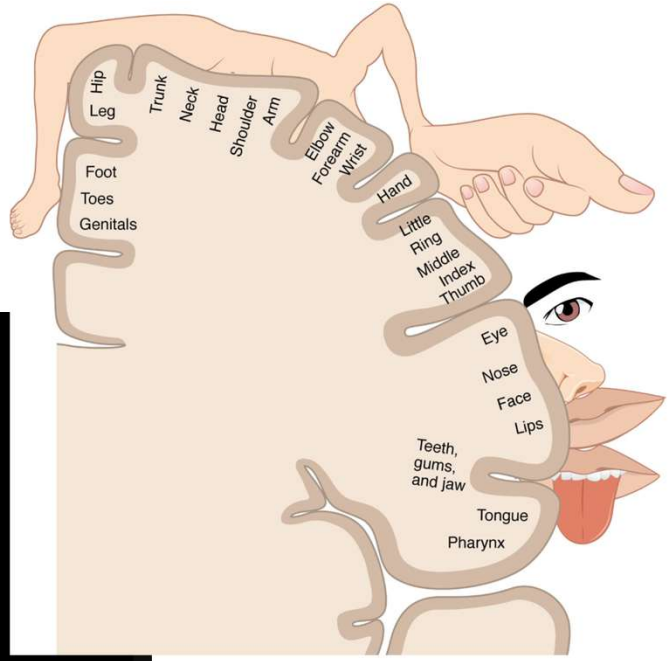
- Animal studies:
  - Medulla oblongata – 3 neuron populations (visceral, somatic and both)
  - Lateral and rostral ventrolateral reticular nuclei + locus ceruleus – somatic
  - Solitary tract nucleus – somatic and visceral
- Human (fMRI)
  - Parabrachial complex, locus ceruleus, red nucleus, ventral tegmental area, substantia nigra, periaqueductal gray matter, cuneiform nucleus – somatic and visceral
  - Obs. resolution



Somatosensorisk cortex

- Somatic homunculus (brains perception of the body)
- Mindre aktivering af S1 (primary somatosensory cortical area) – mere aktivering af insula, ACC, PFC etc. ved visceral smerter
- Dvs. affektiv component ("hvor ubehagelig"), autonomy – mindre præcis lokalisation



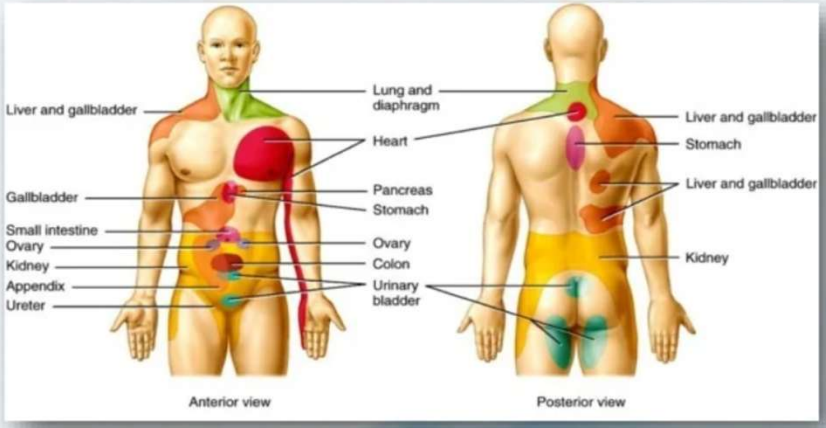


IBS:

- Comprise 50% of referrals to GA
- Affects up to 20% of US population

Examples of Visceral Pain

- Myocardial ischemia[1]
- Kidney and ureteral stones[1]
- Peptic ulcer
- Inflammatory bowel diseases (Crohn's disease[3], Ulcerative colitis)
- Biliary colic/ cholecystitis
- Appendicitis, diverticulitis[6]
- Pancreatitis[3]
- Cancer[3]
- Irritable bowel syndrome[1]
- Functional dyspepsia[3]
- Endometriosis[5]
- Vulvodynia[2]
- Interstitial cystitis[4]



1. ISAP. Visceral Pain. Available at: http://iasp.files.cms-plus.com/Content/ContentFolders/Publications2/PainClinicalUpdates/Archives/PCU05-6_1390263870315_22.pdf. Accessed 5 March, 2015; 2. Wesselmann U *et al. Drug Discov Today Ther Strateg*. 2009;6(3):89-95; 3. Davis MP. *Pain Res Treat*. 2012;2012:265605; 4. Wesselmann U. *Urology*. 2001;57(6 Suppl 1):32-9; 5. Issa B *et al. Gut*. 2012;61(3):367-72; 6. Humes DJ *et al. Neurogastroenterol Motil*. 2012;24(4):318-e163.

Prævalens mavesmerter

En af de hyppigste årsager til henvisning

Table 2. Leading Gastrointestinal Symptoms Prompting an Outpatient Clinic Visit in the United States, 2000

Symptom	Estimated no. of visits[a]
Abdominal pain, cramps, spasms	12,275,373
Diarrhea	
Nausea	3,316,009
Vomiting	2,884,373
Heartburn and indigestion (dyspepsia)	1,824,043
Constipation	1,327,156
Anal/rectal bleeding	1,260,644
Blood in stool (melena)	1,179,224
Other, unspecified gastrointestinal symptoms	957,866
Decreased appetite	715,737
Difficulty swallowing (dysphagia)	676,421
Anal/rectal pain	514,896
Other symptoms or changes in bowel function	466,803
Jaundice	431,410
Abdominal distension, fullness, or swelling	384,877
Abdominal mass or tumor	247,850
Gastrointestinal bleeding	229,729
Anal/rectal swelling or mass	224,093
Flatulence	209,434

NOTE. Source: National Ambulatory Medical Care Survey 2000 (http://www.cdc.gov/nchs/nhcs.htm).
[a]The total number of outpatient visits was estimated from a random sample of 27,369 patient visits to 1388 physicians participating in the National Ambulatory Medical Care Survey 2000.

12.275.373

GASTROENTEROLOGY 2004;126:1448–1453

Digestive and Liver Diseases Statistics, 2004

MARK W. RUSSO, JEFFREY T. WEI, MICHELLE T. THINY, LISA M. GANGAROSA, ALPHONSO BROWN, YEHUDA RINGEL, NICHOLAS J. SHAHEEN, and ROBERT S. SANDLER
Division of Gastroenterology and Hepatology, Department of Medicine, and the Center for Gastrointestinal Biology and Disease, University of North Carolina, Chapel Hill, North Carolina

Background & Aims: Digestive and liver diseases are associated with substantial morbidity and mortality in the United States. Statistics about the incidence, prevalence, mortality, and resource utilization of digestive and liver diseases in the United States may be cumbersome to obtain because they are scattered in multiple sources. These data may be useful for policy makers, grant applicants, and authors. **Methods:** Data on the most common gastrointestinal and liver diseases were collected from large publicly available national databases. Information was collected on inpatient and outpatient gastrointestinal complaints and diagnoses, gastrointestinal cancers, and deaths from common liver diseases. **Results:** The leading gastrointestinal complaint prompting an outpatient visit is abdominal pain, with 12.2 million annual visits, followed by diarrhea, nausea, and vomiting. Abdominal pain is the leading outpatient gastrointestinal diagnosis, accounting for 5.2 million visits annually, followed by gastroesophageal reflux disease, with 4.5 million visits. Gallstone disease is the most common inpatient diagnosis, with 262,411 hospitalizations and a median inpatient charge of $11,584. Colorectal cancer is the most common gastrointestinal cause of death and is the most common gastrointestinal cancer, with an incidence of 54 per 100,000. Among gastrointestinal cancers, primary liver cancer had the highest increase in incidence from 1992 to 2000. **Conclusions:** Gastrointestinal and liver diseases are associated with significant outpatient and inpatient healthcare utilization. Following trends in utilization is important for determining allocation of resources for health care and research.

Digestive and liver diseases are common in the United States. They affect large numbers of individuals and exact a considerable financial and social burden. These diseases are associated with frequent physician visits and hospitalizations and are sometimes fatal.

Although data on digestive and liver conditions are collected on a regular basis, they are sometimes difficult to locate. Moreover, previous compilations of statistics on digestive diseases are lengthy or do not provide data on outpatients.[1,2] The American Gastroenterological Association publication, *The Burden of Gastrointestinal Disease*, is a comprehensive report, but it is lengthy and has not been recently updated.[1] Herein, we provide a concise description of frequently used statistics displayed in tables comparing common gastrointestinal diagnoses. A publication on selected digestive diseases in the United States focused mostly on costs and does not provide data on symptoms and diagnoses for outpatient visits or on the incidence and prevalence of gastrointestinal cancers.[2]

Our goals were to collect and report the most recent statistics on deaths, symptoms, physician visits, and hospitalizations for common digestive and liver diseases. These data are unique because they are population based and are gathered from large national databases. This information should be a valuable reference for authors, grant applicants, funding agencies, and policy makers. Importantly, the data are likely to raise questions and lead to future research.

Materials and Methods

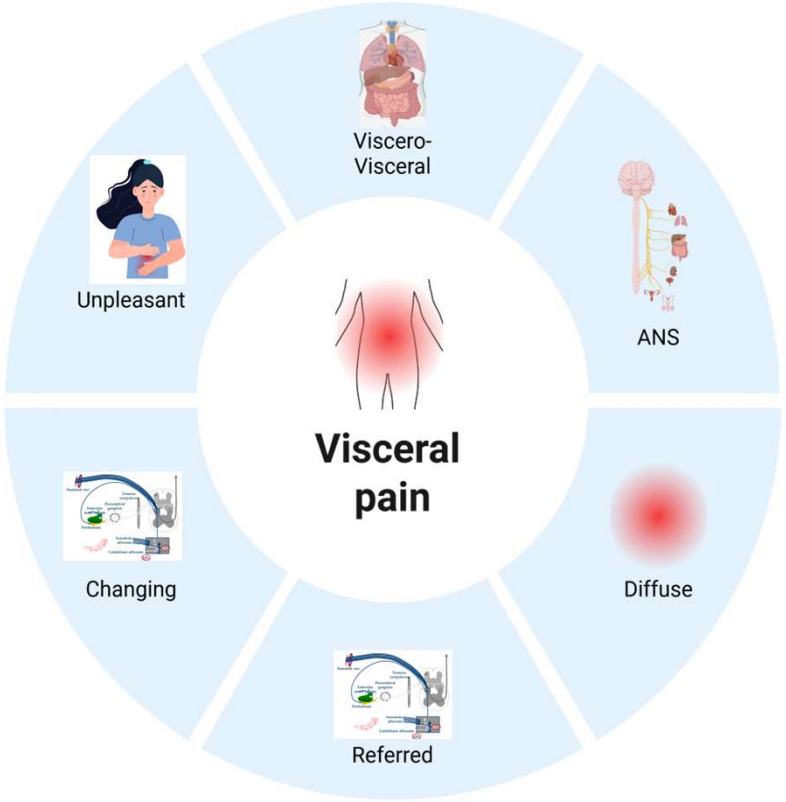
Gastrointestinal Causes of Death

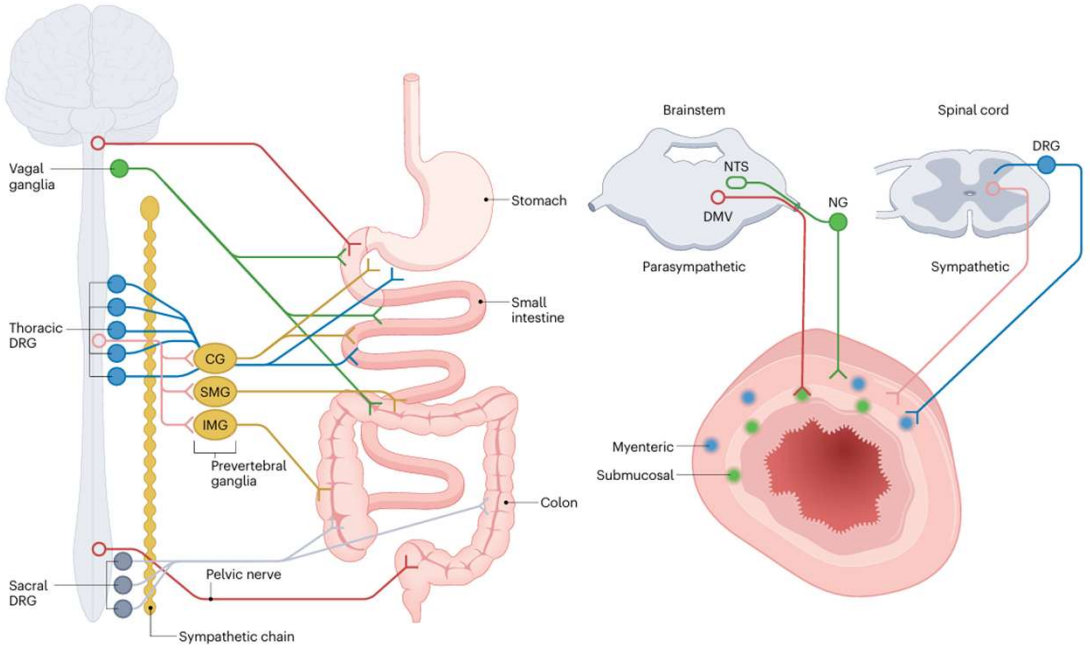
Data on the leading causes of death were obtained from mortality tables from the Division of Vital Statistics, National Center for Health Statistics (Table 1). A total of 2,403,351 deaths in the United States were reported during the year 2000. Causes of death were reported on the basis of the 10th revision of the International Classification of Diseases (ICD-10; 1992). In constructing these tables, the underlying cause of death is ascertained from death certificate data and then coded according to the most recent revision of ICD-10 codes. ICD-10 codes were published by the World Health Organization in 1992 and have been used for coding and classifying causes of death since 1999. ICD-10 codes differ from ICD-9 codes in some ways, although the overall content is similar. ICD-10 has alphanumeric categories rather than numeric categories and

Abbreviations used in this paper: ICD, International Classification of Diseases; NAMCS, National Ambulatory Medical Care Survey; NIS, Nationwide Inpatient Sample; SEER, Surveillance, Epidemiology, and End Results.

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0016-5085/04/$30.00
doi:10.1053/j.gastro.2004.01.025

Visceral pain

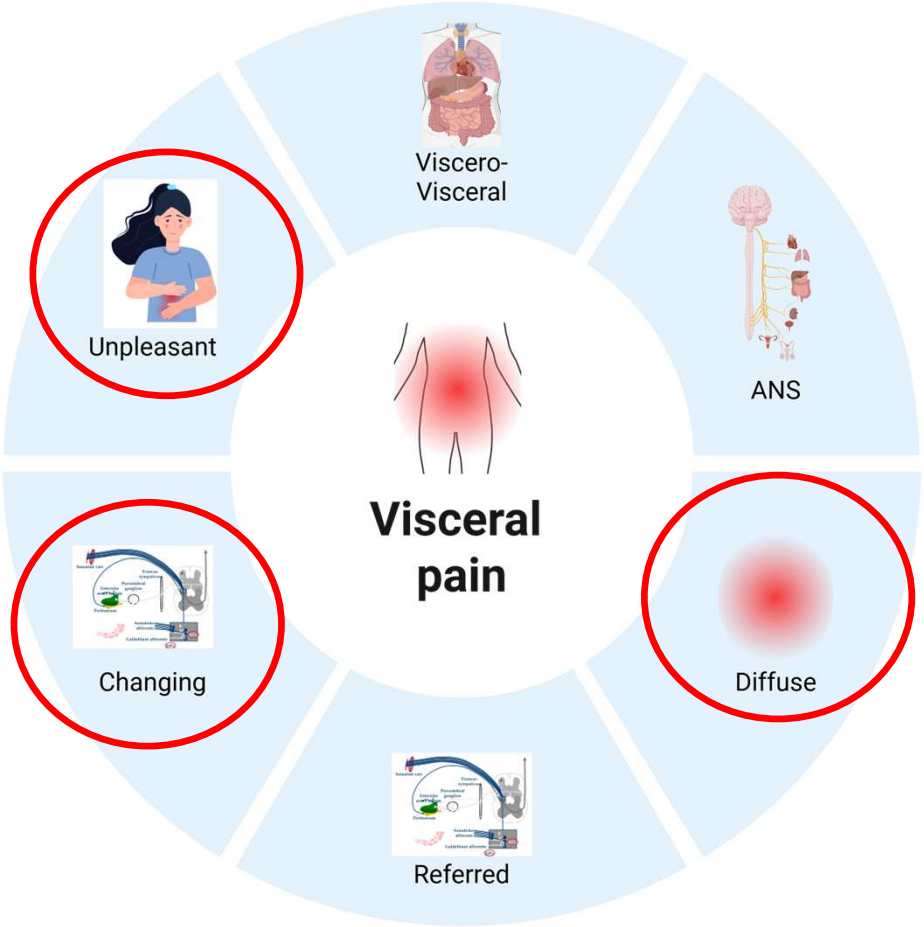




Meerschaert and Chiu 2025. https://doi.org/10.1038/s41575-024-01017-9

Changing

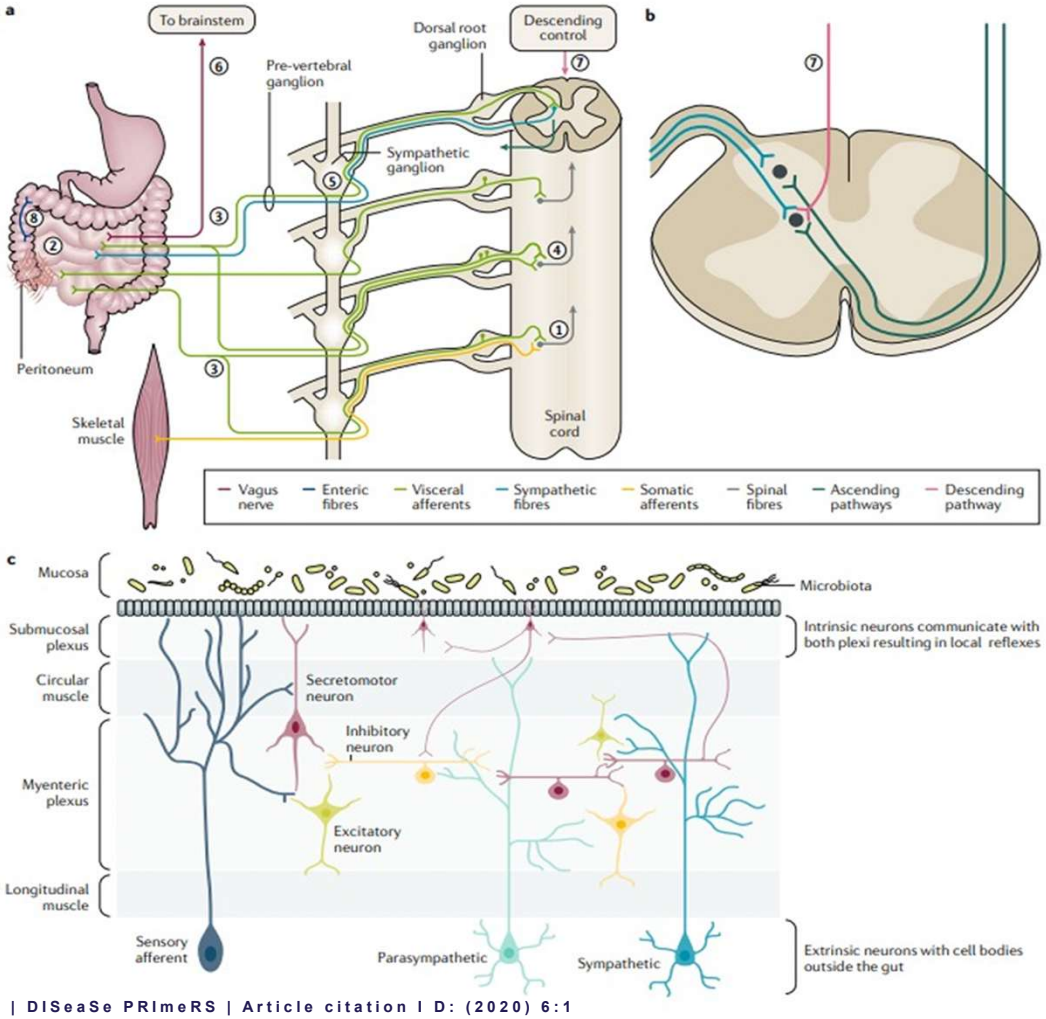
- Smerter ændrer sig:
  - Initial diffuse lokalisering i midtlinjen (primære viscerale afferenter)
  - ANS aktivering (kvalme, svedudbrud etc.)
  - Smerte lokaliseres (aktivering af silent fibre) og muskel innervering
  - Smerter i McBurney´s punkt: Transmural påvirkning af peritoneum (somatisk smerte)



Patofysiologi viscerale smerter

- Kompleks innervering forklarer heterogeniciteten i den klinske præsentation
- Samme “motorvej” – sympatiske nervesystem
- Cross-talk/kommunikation – sympatiske symptomer som sveden, temp. øgning

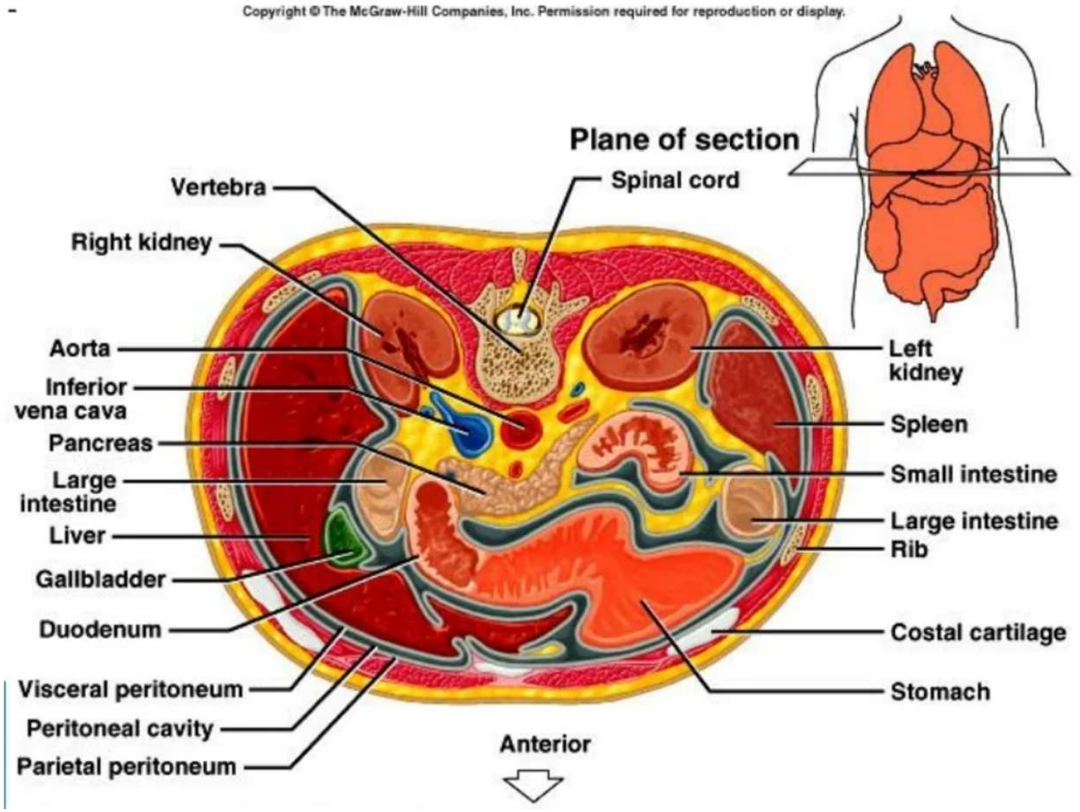
- Klassisk appenticit: ubehag, kvalme, sved etc. (ikke altid så mange smerter i starten)



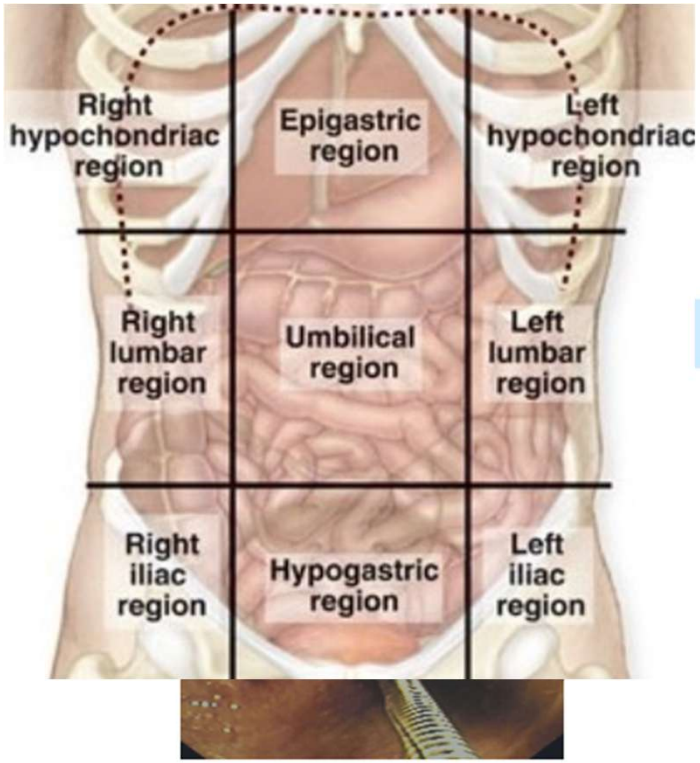
Drewes et al. 2020. Gastrointestinal pain. NATuRe ReVIeWS | DISeaSe PRImeRS | Article citation I D: (2020) 6:1
https://doi.org/10.1038/ s41572-019-0135-7

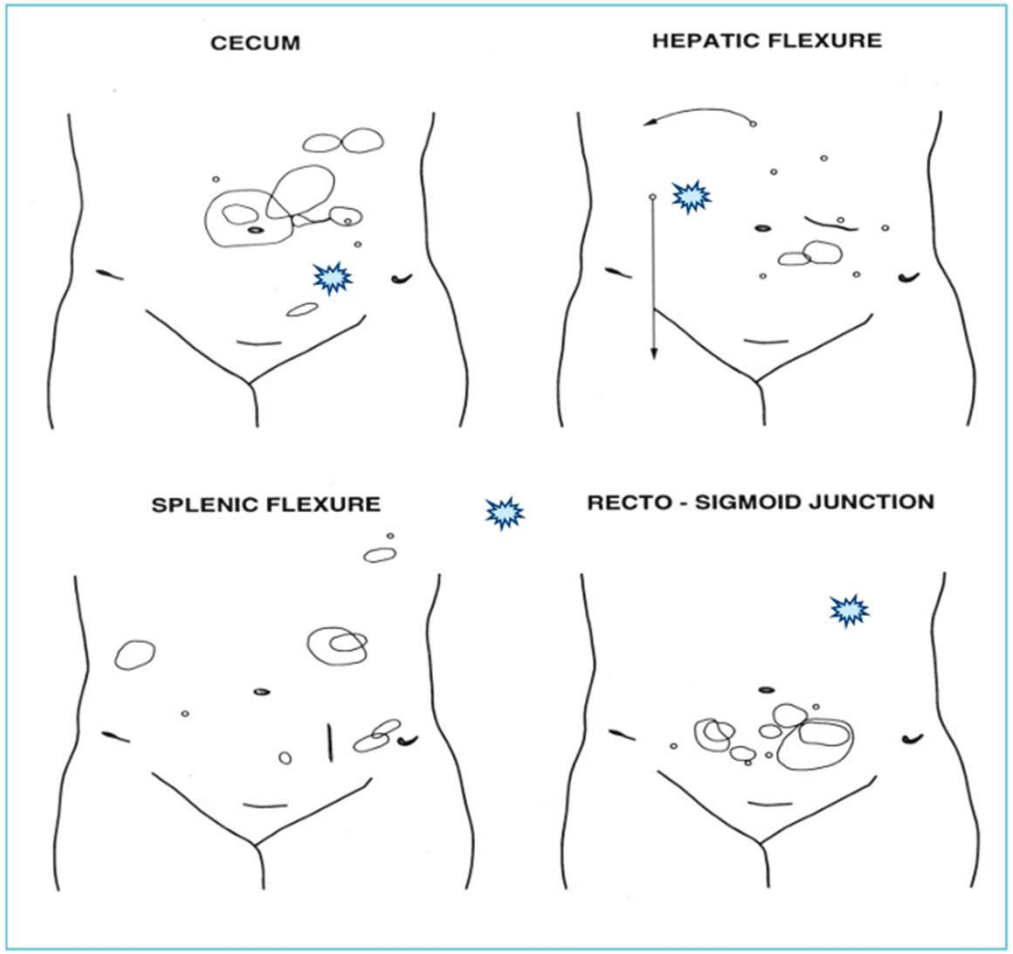
Peritoneum parietale

- Somatisk innervation (ca. 8.uge (embryologisk ophav: mesodermen))
- Præcis lokalisering
- Lubricating fluid – less friction between organs
- Structural support to abdominal organs
- Pathways for nerves, blood vessels and lymphatic vessels
- GI cancer (metastaser i peritoneum)
  - Lign. sympt. som mavesmerter



Ringe lokalisering





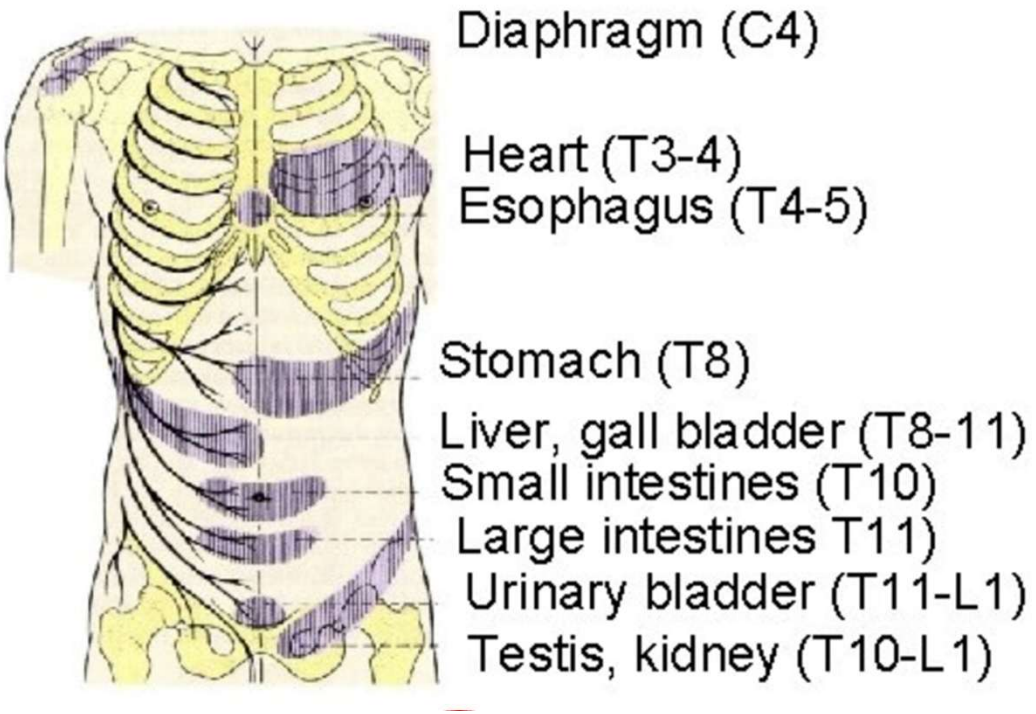
Meddelt smerte

- Typisk undersøgt i dyremodeller
- Eksperimentelle humane smertermodeller
- Sir Henry Head 1893: beskrev viscerotomer (øget sensitivitet i huden pga. visceral sygdom) (Head´s Zones)

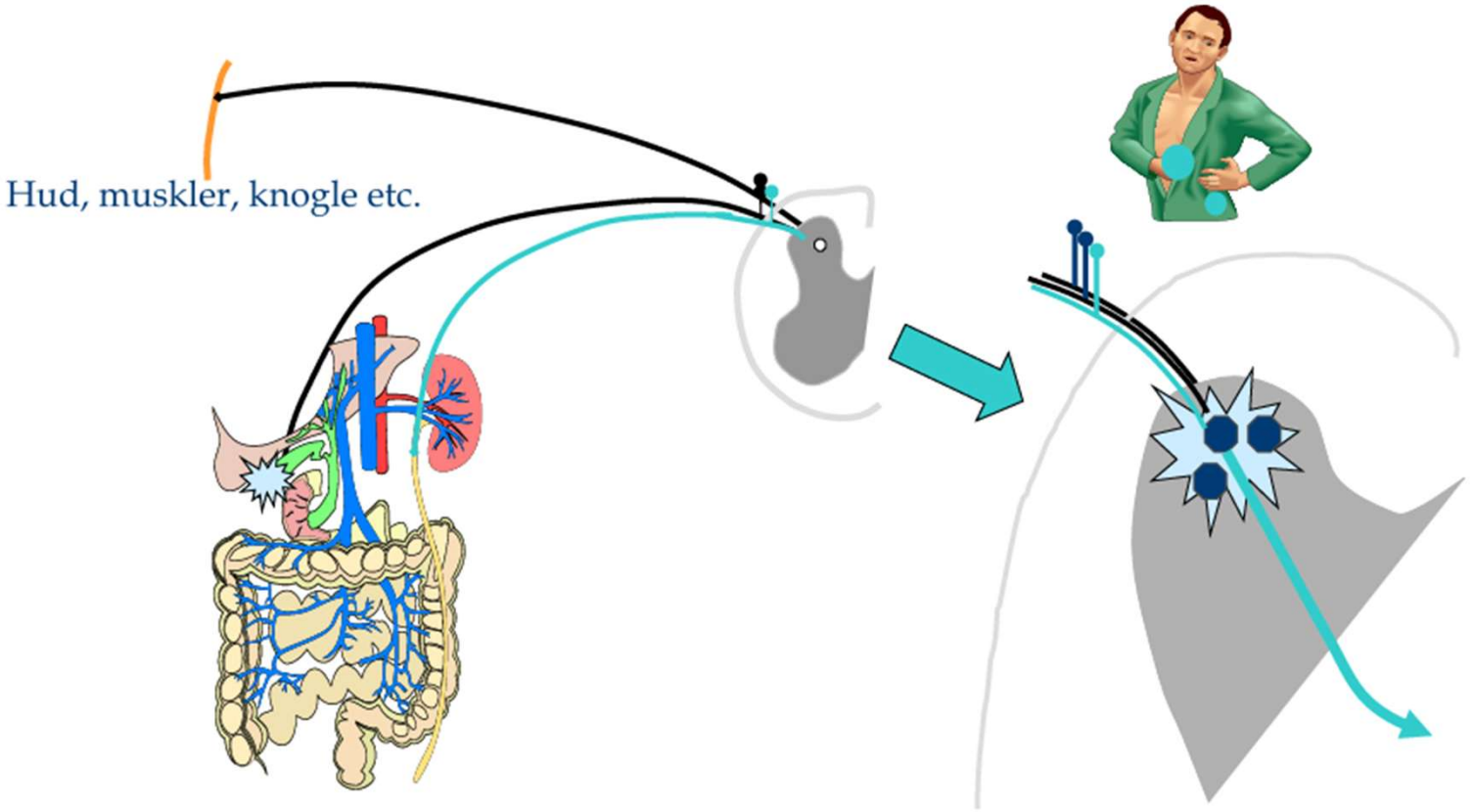
- Klassisk eksempel på meddelt smerte:
  - Smerte i venstre arm pga. angino pectoris

- I mange tilfælde er lokalisationen af meddelt smerter dybt uforudsigelig (Eks. Barrets øsophagus)

- Akut abdomen – CT-skanning: klinisk diagnose ændres i op til 50% af tilfældene

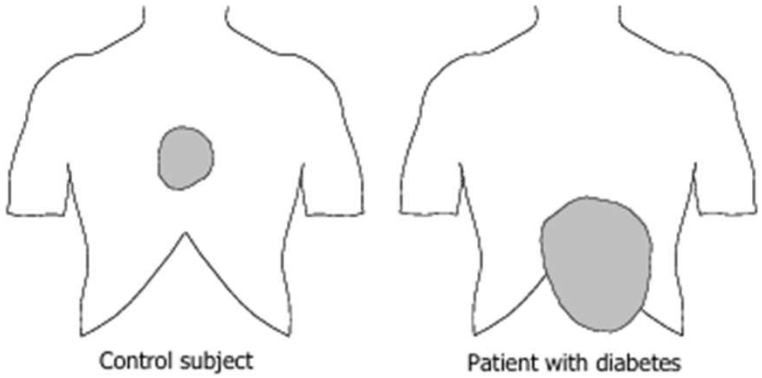


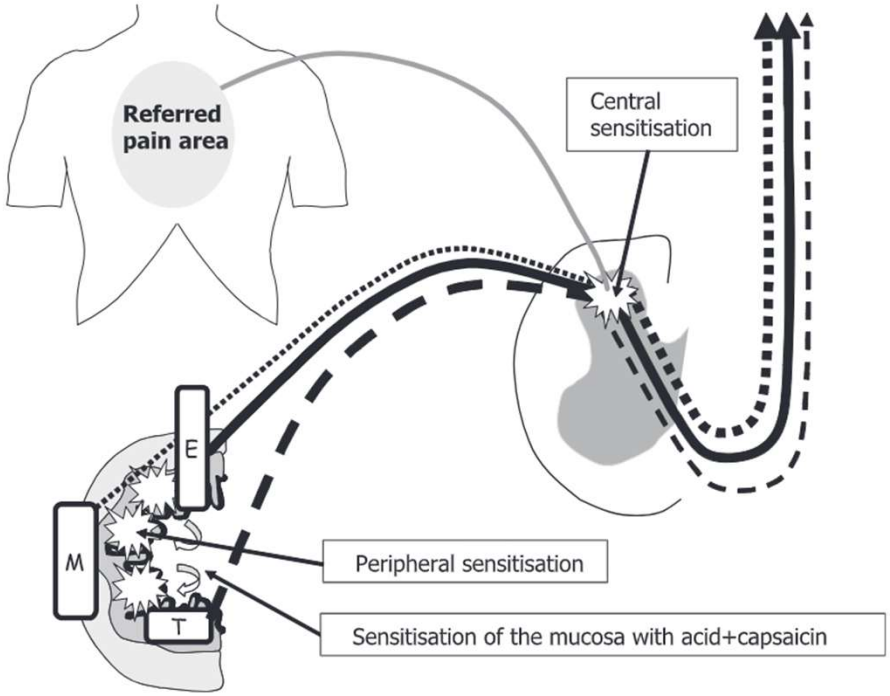
Princip bag meddelt smerte



Eksperimental human model

- Referred pain





Drewes and Gregersen 2006. World J Gastroenterol 2006 April 28; 12(16):2477-2486

Spredning til andre organer

- Perifert niveau
  - Perifer sensibilisering
- Spinalt niveau
  - Hyperalgesi
  - Central sensibilisering
  - NMDA receptor
- Supraspinal niveau
  - Cortikal plasticitet

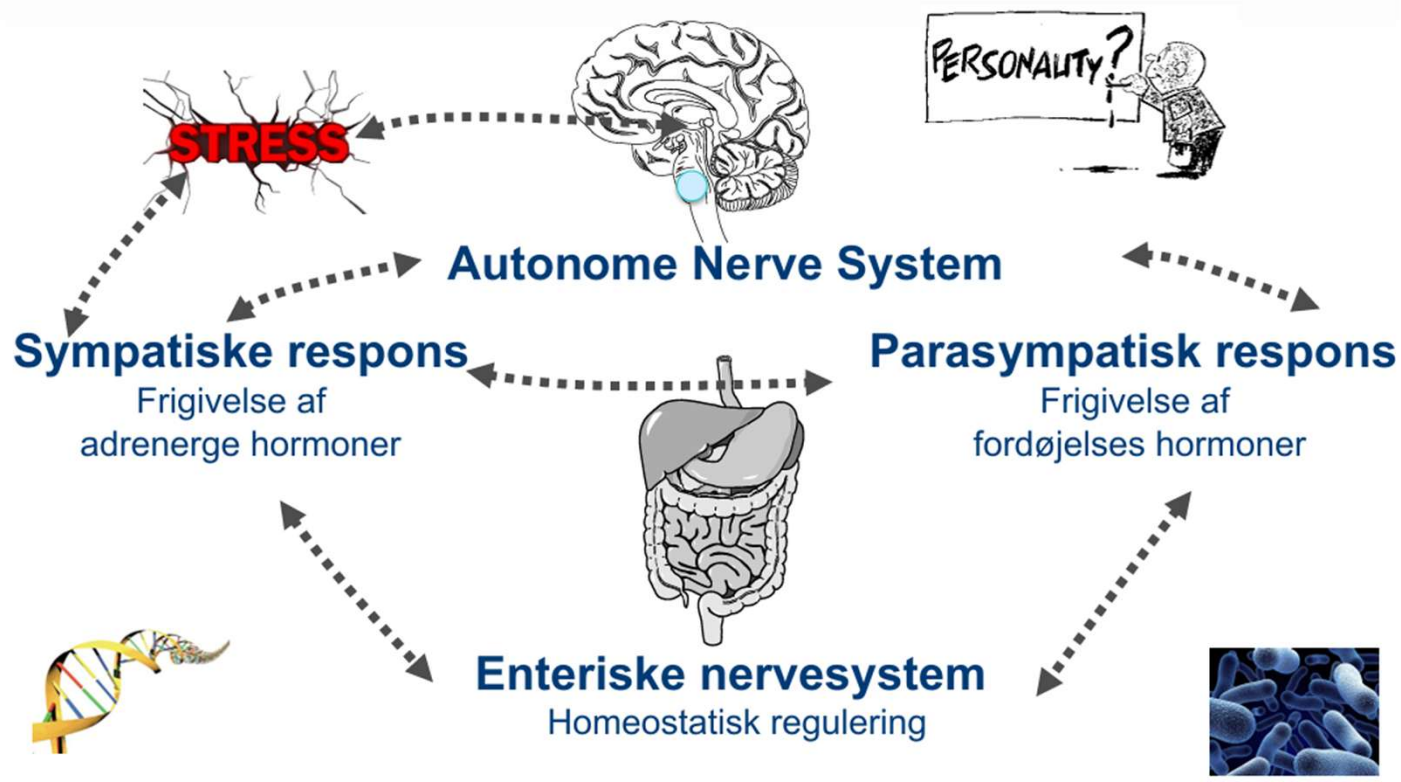
Clinical observations of cross-organ sensitization.

	Organs involved	References
Thoraco-upper abdominal cross-organ sensitization	Lower esophagus and upper esophagus	Sarkar et al. (2000, 2001), Frokjaer et al. (2005), Sarkar et al. (2006)
	Lower esophagus and duodenum	Frokjaer et al. (2005)
	Lower esophagus and rectum	Frokjaer et al. (2005)
	Duodenum and esophagus	Hobson et al. (2004)
	Hypoalgesia of esophageal and duodenal distension in painful chronic pancreatitis patients	Dimcevski et al. (2006)
Pelvic-lower abdominal cross-organ sensitization	Colon and other viscera	Whorwell et al. (1986)
	Bladder and colon	Alagiri et al. (1997)
	Sigmoid colon and rectum	(Naliboff et al., 1997; Lembo et al., 1997; Munakata et al., 1997)
	Reproductive organs and upper urinary system	Giamberardino et al. (2001)

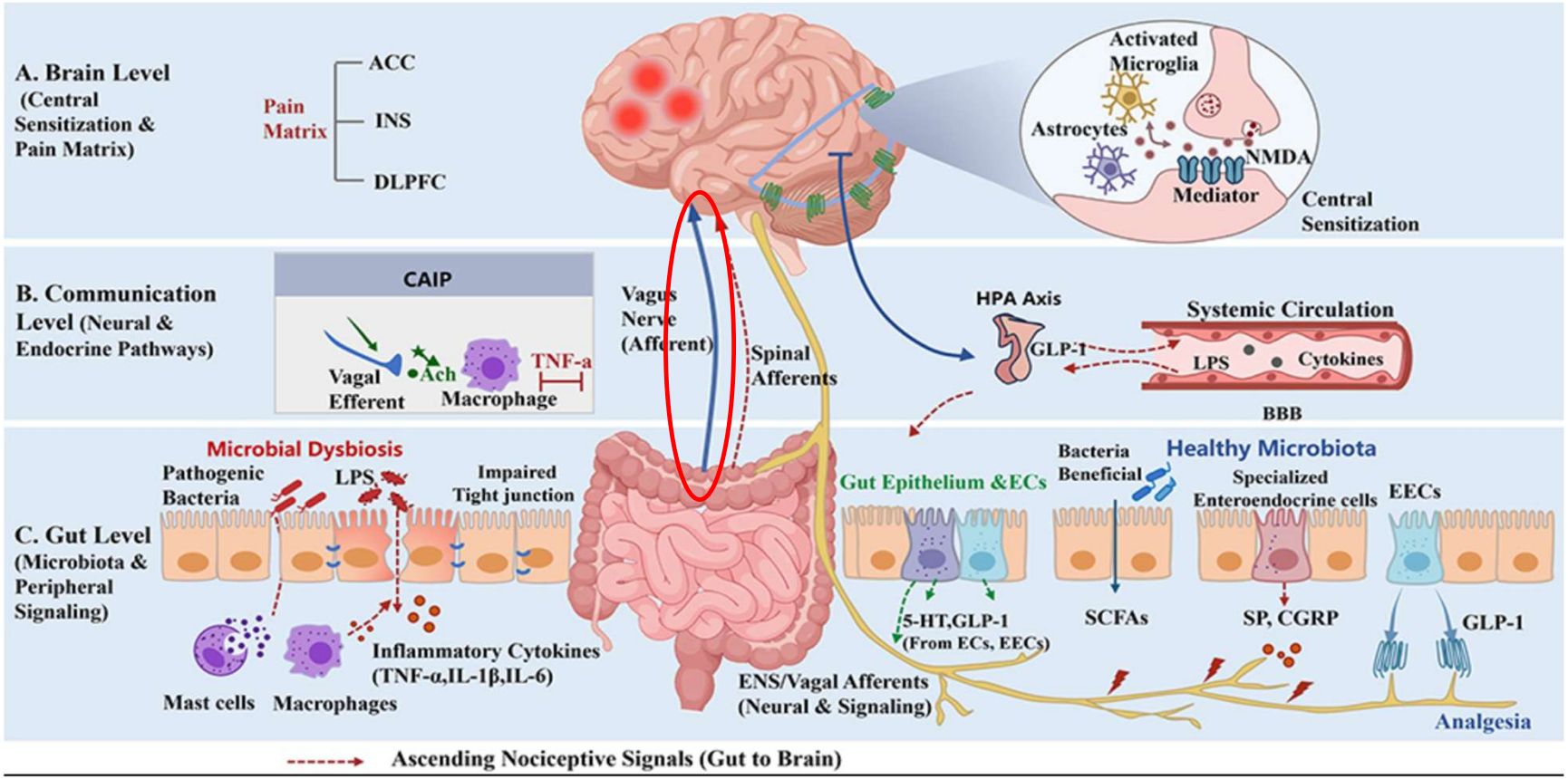
Brumovsky and Gebhart_2010. Visceral organ cross-sensitation – An integrated perspective. Autonomic Neuroscience: Basic and Clinical 153 (2010) 106–115

Brain Gut Axis

- Stress – øget tarmmotilitet, immunaktivitet, visceral smerteføl somhed
- ENS – serotonin, cholecystokinin (fremmer mæthed), GLP-1 (appetit, glukosekontrol)
- Kronisk viscerale smerter
  - Perifer sensibilisering
  - Central sensibilisering
  - Viscero-visceral pain



Brain Gut axis

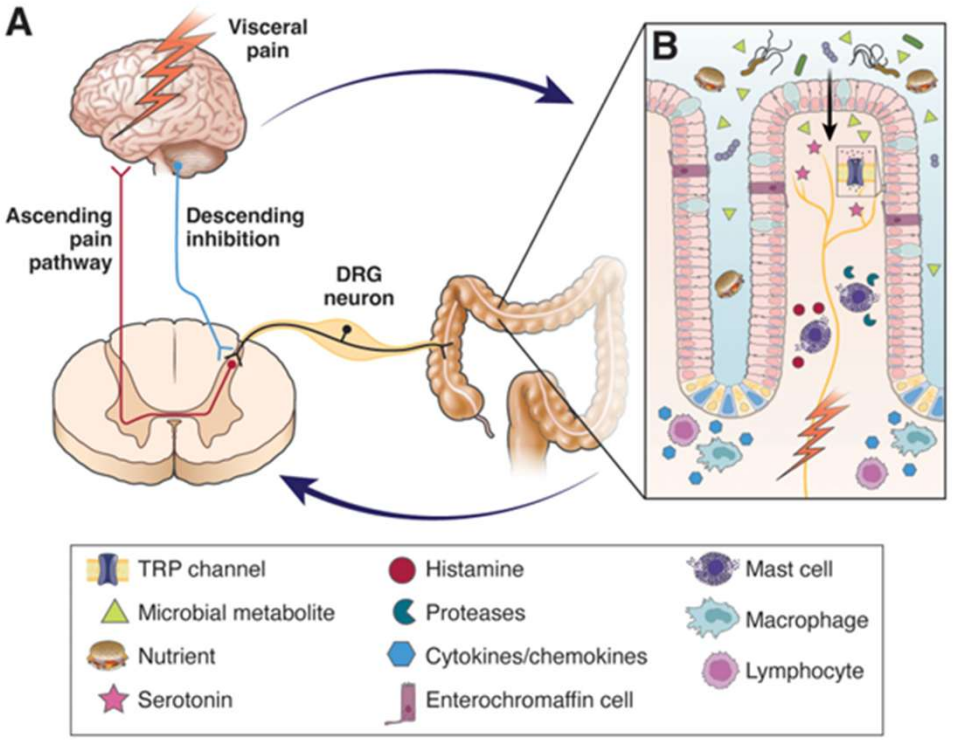


Song et al. 2026. Bidirectional Communication of the Gut-Brain Axis in Pain Regulation: From Microbial Metabolites to Neuroinflammation. Journal of Inflammation Research 2026:19 614847

Peripheral level

- Opregulering af neurotransmitter
  - “Suppe” af inflammatoriske faktorer
  - H+-ioner, noradrenalin, bradykinin, histamine, cytokiner, NGF, neuropeptider etc.
  - Activation of TRP ion channels / G-protein-coupled receptors on nociceptors
- Activation of low-threshold and high-threshold fibers
- Aktivation of silent fibre?

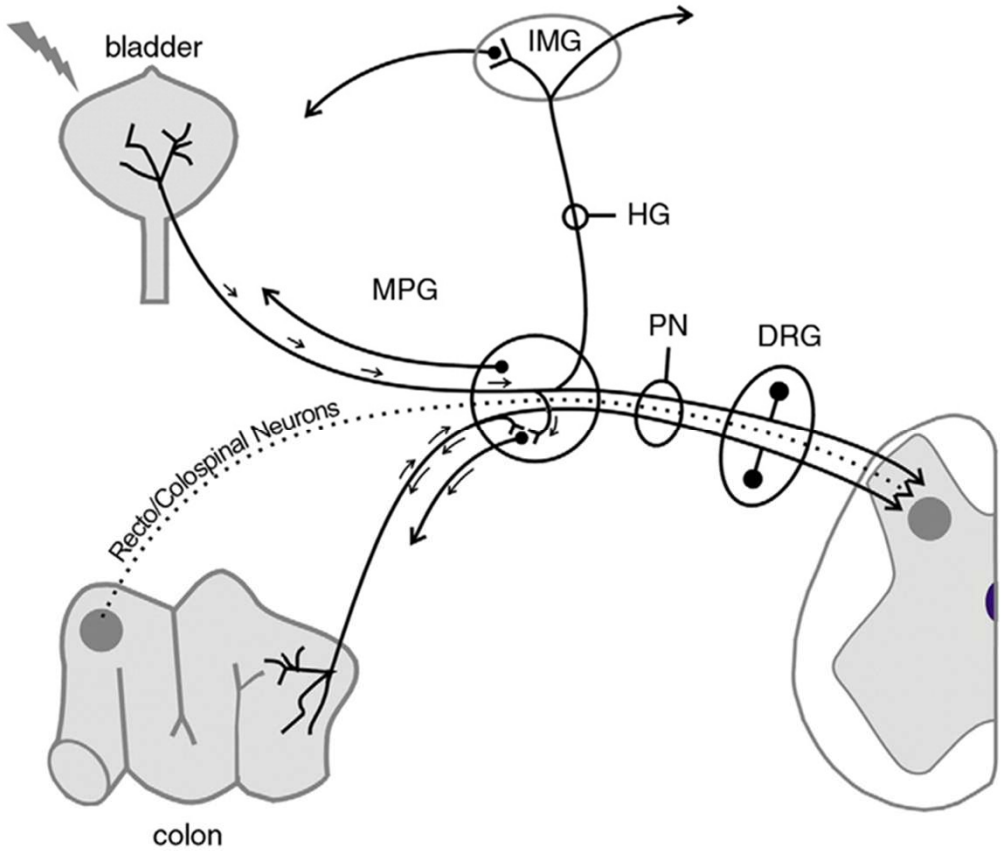
- Increase in peripheral afferent drive – central sensitization



Song et al. 2026. Bidirectional Communication of the Gut-Brain Axis in Pain Regulation: From Microbial Metabolites to Neuroinflammation. Journal of Inflammation Research 2026:19 614847

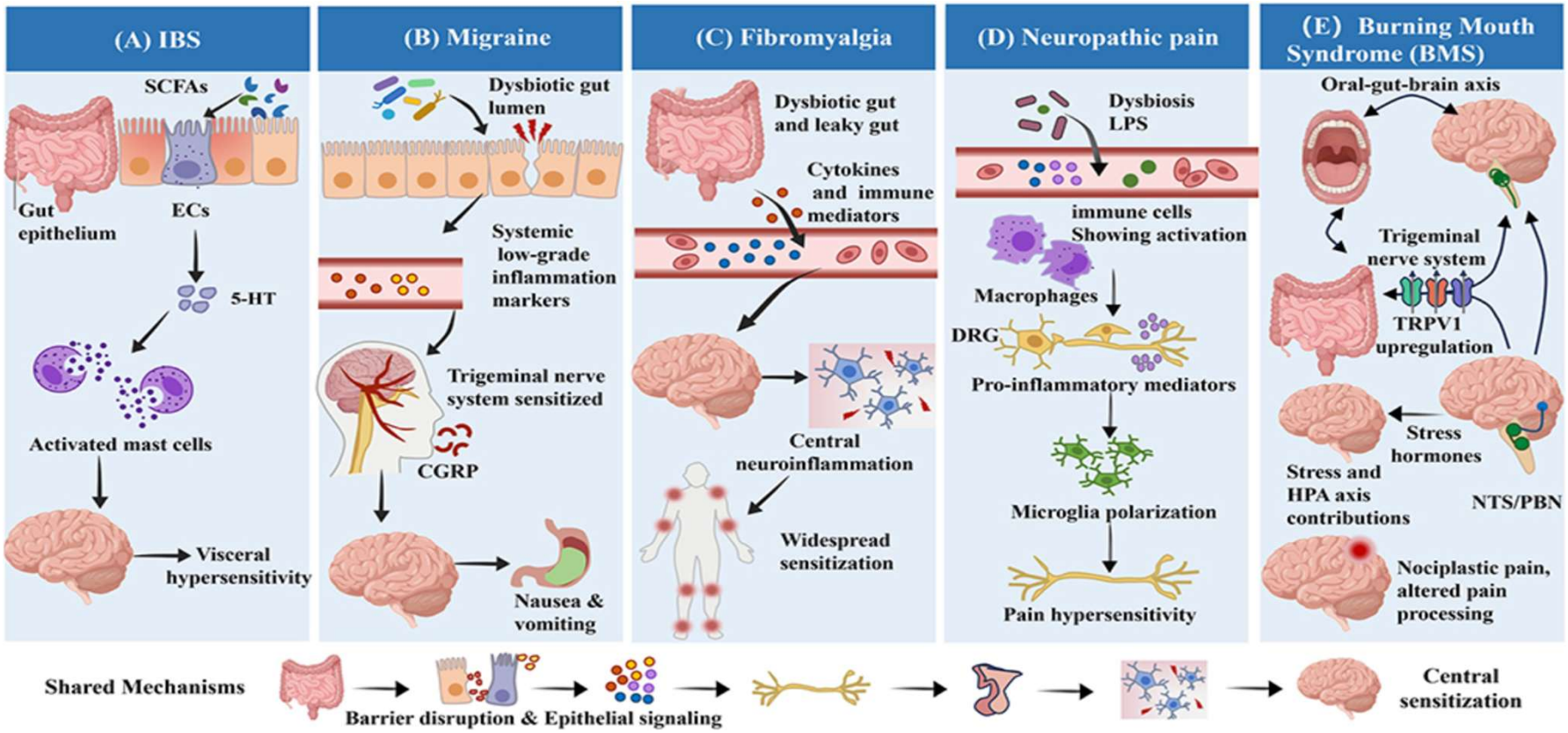
Peripherally mediated

- ANS



Brumovsky and Gebhart 2009. Visceral organ cross-sensitization – An integrated perspective. Autonomic Neuroscience: Basic and Clinical 153 (2010) 106–115

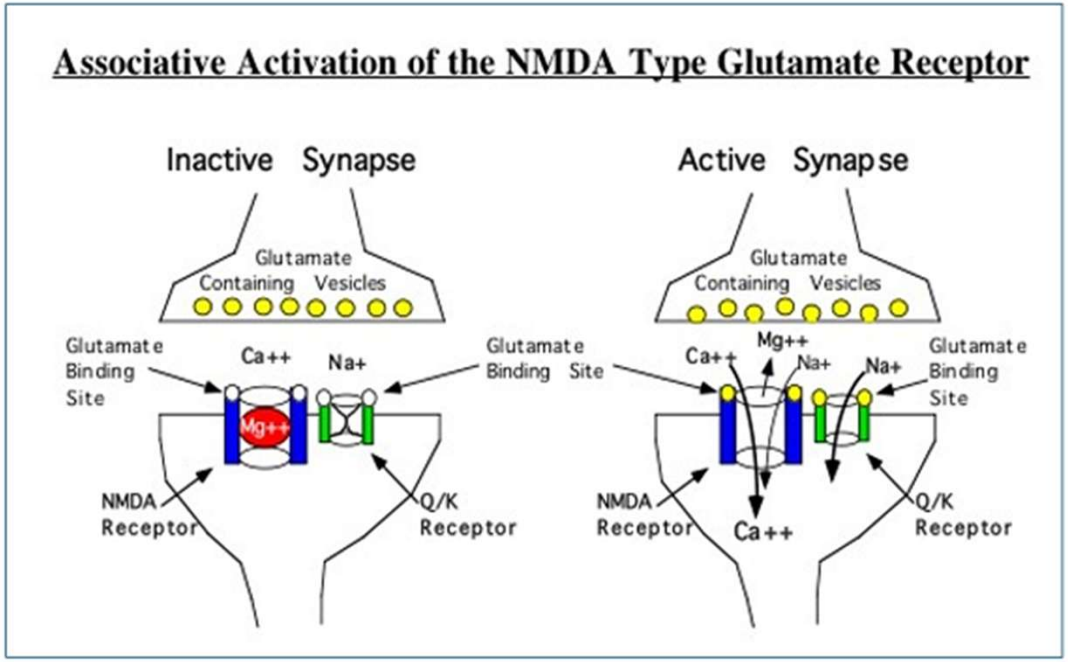
Eksempler



Song et al. 2026. Bidirectional Communication of the Gut-Brain Axis in Pain Regulation: From Microbial Metabolites to Neuroinflammation. Journal of Inflammation Research 2026:19 614847

Nociplastisk Sensibilisering

- Smerte ikke stationært
- Langvarige/kroniske smerter: øget central sensibilisering (forsvinder ikke selvom sygdom forsvinder/mindskes)
- F.eks. akut inflammation i tarmen – op til 17-20% udvikler kolon irritable efterfølgende
- Substance P: hurtigere excitatorisk potentiale
- Neurokinin 1 aktiveres
- NMDA receptor
- Protein kinase A og C, Nitrogenoxid synthetase opreguleres
- Tab af inhiberende interneuroner, tab af GABA



Opsummering af symptomer

- Atypisk lokalisering
- Dårlig correlation mellem organisk sygdom (peptisk ulcus, blindtarmsbetændelse) og smerte intensitet
- Aktivering af viscerale autonome reflekser og det enteriske nervesystem
  - Nedsat peristaltic/motilitet
  - Svedudbrud
  - Kvalme
  - Ubehag
  - Opkastning

Klinisk undersøgelse

- SPØRG PATIENTEN
- Hvornår gør det ondt/oplever du symptomer f.eks. post prandialt?
- Anamnese:
  - Debut, konstant/intermitterende smerte
- Generel tilstand
  - Bevidsthed, vaskulært shock (bleghed, mikrocirkulation), septikæmi (kold-svedende)
- Akut abdomen
  - Peritoneal involvering – fysisk bevægelse forværrer ofte smerteoplevelsen
- INSPEKTION
- Tidligere abdominal kirurgi
- Abdominal respiration
- Misfarvning af husen (blødning fra milt, aorta aneurism)
- Udvidelse af bugomfanget

Klinisk undersøgelse

- AUSKULTATION
- Karakteriser tarmbevægelserne/peristaltik
- Lyt kontinuerligt i 3 min.
- Tavshed er et advarselstegn
- Metalliske lyde også varselstegn



- PERCUSSION
- Luft (f.eks. tarmslyng – hul/trommende lyd)
- Dæmet lyd (f.eks. fibrinudsvedning v. blindtarmsbetændelse)
- Dæmpning over symfysen (urin retention)
- Ydmyghed ved fortolkning – uspecifik (vurder m. andre fund)



Klinisk undersøgelse

- PALPATION
- Differentier om der er involvering af peritoneum eller ej
- Udløser palpitation smerter?
- Indirekte smerter (pressure and release) (slipømhed)
- Defensivt smerteværn

- Visceral smerte: diffuse, dump, ofte uden slipømhed



Opsummering

- Visceral smerte adskiller sig på mange måder fra somatisk smerte
- Involvering af ANS
- Dårlig organotopisk lokalisering
- OBS: meddelt smerte
- Segmental overlap
- Peritoneum parietale modtager somatisk innervation

Box 1 | **Clinical characteristics of GI pain**

- True gastrointestinal (GI) pain is poorly localized and often associated with autonomic symptoms, such as the initial phase of acute appendicitis.
- Referred pain to somatic structures is observed in typical dermatomes according to the affected organ, but may be abnormal in localization and extent, especially in functional diseases.
- Cross-organ sensitization can lead to symptoms in otherwise healthy organs and may complicate diagnosis.
- Involvement of the enteric nervous system is frequent and can result in GI dysmotility that may be symptomatic.
- When a GI disease affects the peritoneum or other visceral membranes, the clinical presentation may shift from typical visceral (diffuse) to somatic (sharp, well-localized) pain.
- GI pain may increase sympathetic and parasympathetic activity and vagal reflexes can, for example, change fluid transport, pancreatic function and cardiac and respiratory rhythm; accordingly, diarrhoea, sweating and palpitations can dominate the clinical picture.
- Sensitization and neuroplastic changes are frequently observed in GI diseases and may explain the chronicity of pain despite normalization of organ function and afferent barrage.

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